

SIMATS ENGINEERING

ASSESSMENT TOOL 4

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 20%

Time Duration: 1 Hour

QUESTIONS: 5 Total Marks: 30

Annexure A

Code of Conduct:

I BALAJI D (Reg. No: 192512368) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Department: ECE

Signature of the student: _Balaji.D_____

Resolution Algorithm Implementation

Question 1 – Rain and Wet Ground

Knowledge Base: 1. If it rains, then the ground becomes wet. 2. It is raining.

Goal: Prove that the ground is wet using the Resolution Algorithm.

Tasks: (1) Represent in propositional logic. (2) Convert to CNF. (3) Negate the goal. (4) Apply Resolution step by step. (5) Show the Empty Clause ($\square$) if proved.

Answer

Let R = It is raining, W = The ground is wet.

Propositional logic: 1. $R \rightarrow W$ 2. R

CNF: 1. $\neg R \vee W$ 2. R

Negate the goal: $\neg W$

Clauses: $\{\neg R \vee W\}$, $\{R\}$, $\{\neg W\}$

Step 1: Resolve $\{R\}$ with $\{\neg R \vee W\} \rightarrow \{W\}$

Step 2: Resolve $\{W\}$ with $\{\neg W\} \rightarrow \square$ (Empty Clause)

Answer: The empty clause is derived, so the goal is proved: the ground is wet.

Question 2 – Student Assignment Submission

Knowledge Base: 1. If a student submits the assignment, then the student receives internal marks. 2. Rahul submitted the assignment.

Goal: Prove that Rahul receives internal marks using the Resolution Algorithm.

Tasks: (1) Write propositional logic expressions. (2) Convert premises to CNF. (3) Negate the goal. (4) Apply Resolution. (5) Conclude.

Answer

Let S = A student submits the assignment, M = The student receives internal marks.

Propositional logic: 1. $S \rightarrow M$ 2. S

CNF: 1. $\neg S \vee M$ 2. S

Negate the goal: $\neg M$

Step 1: Resolve $\{S\}$ with $\{\neg S \vee M\} \rightarrow \{M\}$

Step 2: Resolve $\{M\}$ with $\{\neg M\} \rightarrow \square$ (Empty Clause)

Answer: The goal is proved: Rahul receives internal marks.

Question 3 – Library Membership

Knowledge Base: 1. If a person is a library member, then the person can borrow books. 2. Priya is a library member.

Goal: Prove that Priya can borrow books using the Resolution Algorithm.

Tasks: (1) Represent using propositional logic. (2) Convert each statement to CNF. (3) Negate the goal. (4) Resolve until no new clauses can be generated. (5) State the final conclusion.

Answer

Let L = A person is a library member, B = The person can borrow books.

Propositional logic: 1. $L \rightarrow B$ 2. L

CNF: 1. $\neg L \vee B$ 2. L

Negate the goal: $\neg B$

Step 1: Resolve $\{L\}$ with $\{\neg L \vee B\} \rightarrow \{B\}$

Step 2: Resolve $\{B\}$ with $\{\neg B\} \rightarrow \square$ (Empty Clause). No further clauses can be generated.

Answer: Final conclusion: the goal is proved — Priya can borrow books.

Question 4 – Placement Eligibility

Knowledge Base: 1. If a student clears the aptitude test, then the student is eligible for placement. 2. Arun cleared the aptitude test.

Goal: Prove that Arun is eligible for placement using the Resolution Algorithm.

Tasks: (1) Convert to propositional logic. (2) Write the CNF clauses. (3) Negate the goal. (4) Apply Resolution step by step. (5) Draw the resolution tree and conclude.

Answer

Let A = A student clears the aptitude test, P = The student is eligible for placement.

Propositional logic: 1. $A \rightarrow P$ 2. A

CNF clauses: 1. $\neg A \vee P$ 2. A

Negate the goal: $\neg P$

Step 1: Resolve $\{A\}$ with $\{\neg A \vee P\} \rightarrow \{P\}$

Step 2: Resolve $\{P\}$ with $\{\neg P\} \rightarrow \square$ (Empty Clause)

Resolution tree: $A \ \& \ (\neg A \vee P) \rightarrow P$; $P \ \& \ \neg P \rightarrow \square$

Answer: Conclusion: the goal is proved — Arun is eligible for placement.

Question 5 – Access Control System

Knowledge Base: 1. If a user enters the correct password, then the user is authenticated. 2. If a user is authenticated, then the user is granted access. 3. The user entered the correct password.

Goal: Prove that the user is granted access using the Resolution Algorithm.

Tasks: (1) Represent in propositional logic. (2) Convert all implications to CNF. (3) Negate the goal. (4) Apply Resolution to derive the Empty Clause ($\square$). (5) Explain each resolution step and write the final conclusion.

Answer

Let C = The user enters the correct password, $Auth$ = The user is authenticated, G = The user is granted access.

Propositional logic: 1. $C \rightarrow Auth$ 2. $Auth \rightarrow G$ 3. C

CNF: 1. $\neg C \vee Auth$ 2. $\neg Auth \vee G$ 3. C

Negate the goal: $\neg G$

Clauses: $\{\neg C \vee Auth\}$, $\{\neg Auth \vee G\}$, $\{C\}$, $\{\neg G\}$

Step 1: Resolve $\{C\}$ with $\{\neg C \vee Auth\} \rightarrow \{Auth\}$ (the user is authenticated, since the password was correct)

Step 2: Resolve $\{Auth\}$ with $\{\neg Auth \vee G\} \rightarrow \{G\}$ (an authenticated user is granted access)

Step 3: Resolve $\{G\}$ with $\{\neg G\} \rightarrow \square$ (Empty Clause) — a direct contradiction with the negated goal

Answer: Final conclusion: the empty clause is derived, proving the goal — the user is granted access.

RUBRICS FOR CALCULATION — Numerical Problems

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation